



147634-003



Injection

1% w/v sterile solution for ivermectin

For the treatment and control of internal parasites of cattle and sheep.

INTRODUCTION

IVOMEC is the breakthrough injectable parasiticide for cattle and sheep. It effectively controls internal parasites that impair livestock health and productivity. Its convenience, broad-spectrum efficacy and wide safety margin make it an ideal product for parasite control in livestock.

PRODUCT DESCRIPTION

IVOMEC is a ready-to-use 1.0% w/v sterile solution of ivermectin. Ivermectin is derived from the avermectins, a family of highly active, broad-spectrum antiparasitic agents which are isolated from fermentation of the soil organism *Streptomyces avermitilis*.

IVOMEC is formulated to deliver the recommended dose level of 200 mcg/kg bodyweight.

Description: Clean, clear pale straw-colored liquid free from visible evidence of contamination.

INDICATIONS

Indicated for the effective treatment and control of the following parasites of cattle and sheep:

CATTLE

PARASITE	Adult	L4	Inhibited L4
Gastrointestinal Roundworms			
<i>Ostertagia lyrata</i>	•	•	
<i>Ostertagia ostertagi</i>	•	•	•
<i>Cooperia oncophora</i>	•	•	
<i>Cooperia pectinata</i>	•	•	
<i>Cooperia punctata</i>	•	•	
<i>Haemonchus placei</i>	•	•	
<i>Trichostrongylus axei</i>	•	•	
<i>Trichostrongylus colubriformis</i>	•	•	
<i>Bunostomum phlebotomum</i>	•	•	
<i>Oesophagostomum radiatum</i>	•	•	
<i>Strongyloides papillosus</i>	•		
<i>Nematodirus helvetianus</i>	•		
<i>Nematodirus spathiger</i>	•		
<i>Trichuris</i> spp.	•		
Lungworms			
<i>Dictyocaulus viviparus</i>	•	•	
Eye Worms			
<i>Thelazia</i> spp.	•		
Warbles			
<i>Hypoderma bovis</i>			
<i>H. lineatum</i>			
Mange Mites			
<i>Psoroptes ovis</i>			
<i>Sarcoptes scabiei</i> var. <i>bovis</i>			
Sucking Lice			
<i>Linognathus vituli</i>			
<i>Haematopinus eurysternus</i>			
<i>Solenopotes capillatus</i>			

The product may also be used as an aid in the control of biting lice (*Damalinea bovis*) and the mange mite *Chorioptes bovis*, but complete elimination may not occur.

Persistent Activity

Given at the recommended dosage of 1 ml per 50 kg bodyweight, the product controls re-infection with the following nematodes up to the duration shown:

Parasite	No. of Days After Treatment
Barbers pole worm - <i>Haemonchus placei</i>	14
Small intestinal worm - <i>Cooperia</i> spp.	14
Hairworm – <i>Trichostrongylus axei</i>	14
Brown stomach worm - <i>Ostertagia ostertagi</i>	21
Nodular worm - <i>Oesophagostomum radiatum</i>	21
Lungworm – <i>Dictyocaulus viviparus</i>	28

To obtain optimal benefit from the persistent activity for grazing animals, it is recommended that calves which are set-stocked in the first grazing season should be treated 3, 8 and 13 weeks after the day of turn-out. This can protect the animals from parasitic gastroenteritis and lungworm disease throughout the grazing season, provided they are set-stocked, all the calves are included in the programme and that no untreated cattle are added to the pasture.

Treated animals should continue to be monitored according to good

husbandry practices.

SHEEP

PARASITE	Adult	L4	Inhibited L4
Gastrointestinal Roundworms			
<i>Ostertagia circumcincta</i>	•	•	•
<i>O. trifurcata</i>	•	•	
<i>Haemonchus contortus</i>	•	•	•
<i>Trichostrongylus axei</i>	•		
<i>T. colubriformis</i>	•	•	
<i>T. vitrinus</i>	•		
<i>Cooperia curticei</i>	•	•	
<i>Oesophagostomum columbianum</i>	•	•	
<i>O. venulosum</i>	•		
<i>Nematodirus filicollis</i>	•	•	
<i>Chabertia ovina</i>	•	•	
<i>Trichuris ovis</i>	•		
Lungworms			
<i>Dictyocaulus filaria</i>	•	•	
<i>Protostrongylus rufescens</i>	•		
Nasal Bots			
<i>Oestrus ovis</i>			
Mange Mites			
<i>Psoroptes ovis</i> *			
Benzimidazole-resistant strains of <i>Haemonchus contortus</i> and <i>Ostertagia circumcincta</i> are also controlled.			

DOSAGE AND ADMINISTRATION

IVOMEC Injection may be given with any standard automatic or single-dose syringe.

CATTLE

The recommended dose volume is 1 ml (10 mg ivermectin) per 50 kg bodyweight which should be given only by subcutaneous injection. Using sterile instruments inject IVOMEC under the loose skin in front of or behind the shoulder. Dosage is according to the following schedule:

Body Weight (kg)	Dose volume (ml)
Up to 50	1.0
51–100	2.0
101–150	3.0
151–200	4.0
201–250	5.0
251–300	6.0
301–350	7.0
351–400	8.0

For cattle weighing over 400 kg calculate the dose at the rate of 1 ml per 50 kg bodyweight

SHEEP

The recommended dose volume is 0.5 ml per 25 kg bodyweight, which should be given only by subcutaneous injection. An acceptable injection site is the neck of the sheep. In woolly sheep, be certain that the needle has penetrated the wool and skin before delivering the dose. As with any injection, aseptic technique should be used.

Body Weight (kg)	Dose volume (ml)
Up to 5	0.1
5.1–10	0.2
10.1–15	0.3
15.1–25	0.5
25.1–50	1.0
50.1–75	1.5
75.1–100	2.0

For sheep weighing over 100 kg calculate the dose at the rate of 0.5 ml per 25 kg.

When treating sheep of less than 16 kg, seek veterinary advice regarding the use of 1 ml disposable syringes graduated in increments of 0.1 ml. For the treatment of individual sheep, a syringe not exceeding 2.0 ml and calibrated in increments of 0.1 ml should be used.

Each ml contains 10 mg of ivermectin sufficient to treat 50 kg of bodyweight of cattle and sheep. The injection may be given with any standard automatic or single-dose or hypodermic syringe. Use of a sterile 17 gauge x ½ inch needle is suggested. Replace with a fresh sterile needle after every 10 to 12 animals. Injection of wet or dirty animals is not recommended.

For the treatment and control of sheep scab, two injections with a seven day interval are required to treat clinical signs of scab and to eliminate mites.

PHARMACODYNAMICS

Ivermectin is a member of the macrocyclic lactone class of endectocides, which have a unique mode of action. Compounds of the class bind selectively and with high affinity to glutamate-gated chloride ion channels which occur in invertebrate nerve and muscle cells. This leads to an increase in the permeability of the cell membrane to chloride ions with hyperpolarization of the nerve or muscle cell, resulting in paralysis and death of the parasite. Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA).

147634-003

The margin of safety for compounds of this class is attributable to the fact that mammals do not have glutamate-gated chloride channels, the macrocyclic lactones have a low affinity for other mammalian ligand-gated chloride channels, and macrocyclic lactones do not readily cross the blood-brain barrier.

PHARMACOKINETICS

Maximum plasma concentration

Cattle

At a dose level of 0.2 mg ivermectin per kg a maximum plasma concentration of 35-50 ng/ml is reached in about 2 days and the half-life in plasma is 2.8 days. It is also established that ivermectin is carried mainly in the plasma (80%). This distribution between plasma and blood cells remains relatively constant.

Sheep

At a dose level of 0.3 mg ivermectin per kg an average peak of 16 ng/ml is reached one day after injection.

Excretion: length of time and route

Cattle

A liquid chromatographic method with fluorescence detection allows the determination of ivermectin residues in tissues. After an injection of 0.3 mg ivermectin per kg, the liver (target tissue) had residues ranging from 454 ppb at 2 days post treatment to 11 ppb at 28 days post treatment. All other tissues had lower residues at all time periods: fat > kidney > muscle.

The injection site had residues shortly after treatment, ranging up to 69 ppm at 2 days withdrawal, but by 28 days the average residue was negligible (< 2 ppb). Cattle receiving a single dose of tritium-labelled ivermectin (0.2 - 0.3 mg/kg body weight) were slaughtered at 7, 14, 21 and 28 days after dosing.

Composites of faeces collected during the first 7 days after dosing contained almost all the dosed radioactivity. Only about 1-2 % of the dosed radioactivity was excreted in the urine.

Analyses of the faeces showed that about 40-50% of the excreted radioactivity was present as unaltered drug. The remaining 50-60% was present as metabolites or degradation products almost all which were more polar than the ivermectin.

Sheep

A liquid chromatographic method with fluorescence detection allows the determination of ivermectin residues in tissues. After an injection of 0.3 mg ivermectin per kg, the liver (target tissue) had residues ranging from 160 ppb at 3 days post treatment to 7.2 ppb at 28 days post treatment. The highest residue levels were recovered in fat (from 230 ppb at 3 days post treatment to 13 ppb at 28 days post treatment). Residues in all tissues were below 30 ppb at 28 days post treatment. Radioactive ivermectin was administered to sheep at a dose rate of 0.3 mg per kg. Analyses of the faeces showed that about 99% of the drug and its metabolites are excreted in the faeces, about 1% being excreted in the urine.

CONTRA-INDICATIONS

Sheep scab (*Psoroptes ovis*) is an extremely contagious external parasite of sheep. To ensure complete control great care must be taken to avoid re-infestation, as mites may be viable for up to 15 days off the sheep. It is important that all sheep which have been in contact with infected sheep are treated. Contact between treated, infected and untreated flocks must be avoided until at least seven days after treatment.

In sheep, treatment of psoroptic mange (sheep scab) with one injection is not recommended because, although clinical improvement may be seen, elimination of all mites may not occur.

This product is not for intravenous or intramuscular use.

The product is specifically for use in the target species. It should not be used in other species as severe adverse reactions, including fatalities in dogs, may occur.

WARNINGS AND PRECAUTIONS

Special warnings for each target species

None identified.

Special precautions for use:

i. Special precautions for use in animals

When treating groups of animals use only an automatic dosing device (with vented draw off apparatus when using the 50 ml vial).

Syringes must be filled from the vial through a dry sterile draw-off needle that has been placed in the vial stopper. Vial stoppers must not be breached more than 20 times.

This product does not contain any antimicrobial preservative.

Swab septum before removing each dose. Use a sterile needle and syringe.

ii. Special precautions to be taken by the person administering the medicinal product to the animals

Do not smoke, drink or eat while handling the product.

Wash hands after use.

Take care to avoid self injection: the product may cause local irritation and/or pain at the site of injection.

INTERACTIONS WITH OTHER MEDICAMENTS

IVOMEC Classic Injection for Cattle and Sheep has been used concurrently without adverse effects with foot and mouth disease vaccine or clostridial vaccine, given at separate injection sites.

Adequate vaccination of sheep against clostridial infections is strongly recommended.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

IVOMEC Classic Injection for Cattle and Sheep can be administered to cows and ewes at any stage of pregnancy or lactation provided that the milk is not intended for human consumption. It can be used in breeding ewes, rams, bulls and cows without affecting fertility. IVOMEC Classic Injection can be given to all ages of animals including young calves and lambs.

PACKAGE INFORMATION

IVOMEC is available in two ready-to-use sizes-50 ml and 200 ml.

ENVIRONMENTAL SAFETY

Studies indicate that when ivermectin comes in contact with the soil, it readily and tightly binds to the soil and becomes inactive over time. Drug containers and any residual contents should be disposed of safely (eg, by burying or incinerating) as free ivermectin may adversely affect fish and certain water-borne organisms.

ADVERSE REACTIONS

Transitory discomfort has been observed in some livestock following subcutaneous administration. A low incidence of soft-tissue swelling at the injection site has been observed. Activity suggesting pain, sometimes intense but usually transient, has been observed in some sheep. These reactions disappeared without treatment.

OVERDOSE AND TREATMENT

Cattle

Single doses of 4.0 mg ivermectin per kg (20 x the use level) given subcutaneously resulted in ataxia and depression. No antidote has been identified, however, symptomatic therapy may be beneficial.

Sheep

Dose levels up to 4 mg ivermectin per kg (20 x the use level), given subcutaneously, resulted in ataxia and depression.

MAJOR INCOMPATIBILITIES

No major incompatibility has been identified.

WITHDRAWAL PERIOD

Cattle (meat & offal): 49 days

Cattle (milk): Do not use in cattle producing milk for human consumption. Do not use in non-lactating dairy cows, including pregnant heifers, within 60 days of calving.

Sheep (meat & offal): 37 days

Sheep (milk) Do not use in sheep producing milk for human consumption.

Store below 25°C. Store bottle in carton to protect from light.

Following withdrawal of the first dose, use the product within 12 months.

Keep this and all drugs out of the reach of children.

FOR ANIMAL USE ONLY

Manufactured by:
Boehringer Ingelheim Animal Health do Brasil Ltda.
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REGISTRATION HOLDER

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EDD-hE9ZHT

Code: 147634-003



Cancel And Replace: 147634-002

PRODUCT: IVOMEC INJ 1% - INSERT	COUNTRY: MALAYSIA (MY)	DIMENSIONS	mm = 210 x 300 mm	inches = 000 x 000 mm	PKG-ES = 5010
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Colors Please use the official PANTONE®(P) matching system for an accurate color representation.	Version A: 10 MAY 2010 - 11040 - 1110000141 B: 10 FEB 2010 - 1001 - 1001 change	NOTE FOR THE SUPPLIER:
BLACK CUT	Paulinia G2	

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Marketing approval:

Regulatory approval:

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